

INSTALLATION & SETUP GUIDE

Objective

The Telecom KPI Anomaly Detection Dashboard is an AI-powered application that detects anomalies in telecom network KPI data using the Isolation Forest machine learning algorithm. The application preprocesses telecom data, performs feature engineering, predicts anomalies, and visualizes the results using an interactive Streamlit dashboard.

System Requirements

The following software is required to run the project.

Component	Version
Operating System	Ubuntu 22.04
Python	3.12
Git	Latest version
VS Code	Recommended
pip	Latest version

Prerequisites

Before running the project, ensure that the following are installed:

- Python 3.12
- Git
- Visual Studio Code
- pip
- Internet Connection

Clone the Repository

1. Open the terminal and run:

```
“git clone https://github.com/vibhutichaddha/AI-based-telecom-network-solution.git”
```

2. Move into the project directory:

```
“cd KPI_anomaly_detection_dashboard”
```

3. Create a Virtual Environment

Linux:

```
python3 -m venv venv
```

Activate:

```
source venv/bin/activate
```

4. Install Dependencies

Install all required packages:

```
pip install -r requirements.txt
```

Required Python Packages

Your requirements.txt should include packages like:

- Pandas
- Numpy
- Scikit-learn
- Streamlit
- Plotly
- Joblib
- Matplotlib

Project Structure:

telecom-ai-capstone/

├─ app/

| └─ streamlit_app.py

├─ data/

├─ docs/

| └─ architecture.pdf

| └─ api_documentation.pdf

| └─ installation_guide.pdf

| └─ evaluation_report.pdf

- | └─ learning_log.pdf
- ├─ models/
- ├─ src/
- ├─ screenshots/
- ├─ presentation/
- | └─ final_presentation.pptx
- ├─ README.md
- ├─ requirements.txt
- ├─ LICENSE
- └─ .gitignore

Running the Project

Execute the scripts in the following order.

Step 1: Data Preprocessing

Python src/preprocessing.py

Output: data/processed/telecom_kpi_cleaned.csv

Step 2: Feature Engineering

Python src/feature_engineering.py

Output: data/processed/telecom_kpi_processed.csv

Step 3: Train the Isolation Forest Model

Python src/model.py

Output: models/isolation_forest.pkl

models/scaler.pkl

data/processed/telecom_kpi_with_anomalies.csv

Step 4: Evaluate the Model

Python src/evaluation.py

Output: reports/model_evaluation.txt

Step 5: Hyperparameter Tuning

Python src/hyperparameter_tuning.py

Output: reports/hyperparameter_tuning.csv

Step 6: Launch the Dashboard

Streamlit run app/streamlit_app.py

The dashboard will open automatically in your web browser.

Environment Variables

This project does not require any environment variables because it runs locally and does not use any external APIs or cloud services.

Troubleshooting

Problem:

ModuleNotFoundError

Solution:

Pip install -r requirements.txt

Problem:

FileNotFoundError

Solution:

Verify that the required dataset exists inside the data/processed/ directory and that the file path is correct.

Problem:

No module named streamlit

Solution:

Pip install streamlit

Problem:

Port already in use

Solution:

Run streamlit on another port:

```
streamlit run app/streamlit_app.py --server.port 8502
```

Problem:

Missing required columns

Solution:

Run the scripts in the correct order:

1. preprocessing.py
2. feature_engineering.py
3. model.py
4. evaluation.py

Expected Output

After successful execution, the project will generate:

models/

|__isolation_forest.pkl

|__scaler.pkl

reports/

|__model_evaluation.txt

|__hyperparameter_tuning.csv

Data/processed/

|__telecom_kpi_cleaned.csv

|__telecom_kpi_processed.csv

|__telecom_kpi_with_anomalies.csv

The Streamlit dashboard will display:

- KPI summary
- Throughput trend

- Latency trend
- SINR distribution
- RSRP vs Throughput scatter plot
- Anomaly distribution
- Download results button

Conclusion

- Following this installation guide allows another developer to clone the repository
- Install the required dependencies
- Execute the complete anomaly detection pipeline
- Launch the Streamlit dashboard successfully